

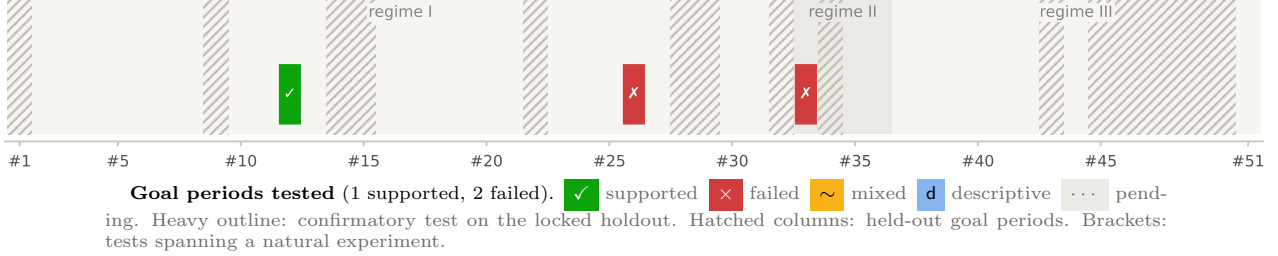
H21: The debate week (#12) is a two-sublattice antiferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2/5

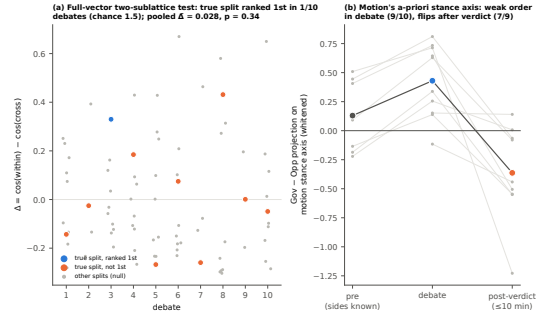
Bottom line. Debaters align on the motion's topic; in reply stance, drafted teams form two sublattices only while assigned, and no camps form without assignment.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Test bed: #12, 10 debates, each with its own judge, motion and re-drafted teams (labels equal DQ6 ground truth). **In content, the pre-registered two-sublattice test fails.** The within- minus cross-team cosine is $\bar{\Delta} = 0.028$ ($p = 0.34$), and content recovers 1/10 teams. **Round 1b** repeats this with a second embedding model and style-residualized and deduplicated inputs ($\bar{\Delta} -0.05$ to $+0.03$). Only a weak tilt along the motion's support-minus-oppose axis remains (bge $p = 0.002$, gte $p = 0.065$). **In reply stance (DQ2), the teams order.** Teammates score $+0.41$ and opponents -0.32 : $\Delta = 0.72$ in 10/10 debates, beyond a calibrated agent-field null ($p = 0.002$). Stance recovers 8/10 teams. The same pair turns hostile only when drafted apart (16/18). The order is gone after the verdict. Where nobody assigns sides (#26 vote, #33 news debate), no camps form. The #34 holdout is not run.



Observables. (a) Per debate, the within-minus-cross cosine for every possible team split (gray) and for the true split (large dot; blue if best). The true split is unremarkable: full content vectors show no two-sublattice order. (b) On the motion's a-priori stance axis, Government sits slightly on the support side during debates (9/10). Post hoc, it swings to the other side after the verdict (7/9).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Vector spins on two sublattices (11), with $q = 2$ Potts (10) and mean field (01). Spin $\mathbf{s}_i$: the masked, whitened ($n = 32$), agent-centred chat mean of debater i in a debate. Sublattice $A = \text{Government}$ ($\varepsilon_i = +1$), $B = \text{Opposition}$. Fields: motion topic $h_u \hat{g}$ (uniform), assigned sides $h_s \hat{a}$ (staggered), family $\mathbf{h}_i$, verdict h_v toward winner W . Couplings: k_{in} (team, ferromagnetic), k_{out} (cross-team, antiferromagnetic). ρ : correlation of sublattice fluctuations.

$$H = -\frac{k_{\text{in}}}{N} \sum_{i \sim j} \mathbf{s}_i \cdot \mathbf{s}_j + \frac{k_{\text{out}}}{N} \sum_{A \times B} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i (h_u \hat{g} + h_s \varepsilon_i \hat{a} + h_v \varepsilon_W \hat{a} + \mathbf{h}_i) \cdot \mathbf{s}_i$$

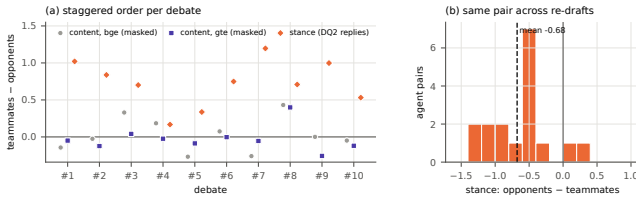
$$\bar{\Delta} = \langle \cos \rangle_{\text{within}} - \langle \cos \rangle_{\text{cross}}, \quad \rho = -k_{\text{out}} / (1 - k_{\text{in}})$$

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RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction	Round 1 $\rightarrow$ round 1b	Verdict
P1 content order	$\bar{\Delta}$ 0.028 $\rightarrow$ $-0.05..+0.03$ (2 models, 5 inputs)	$\times$
P2 teams from content	1/10 $\rightarrow$ 0–1/10	$\times$
P4d text-axis tilt	σ 0.092 (p 0.002) $\rightarrow$ gte 0.055 (p 0.065) $\sim$	
P7 AF fluctuations	ρ +0.53 $\rightarrow$ $+0.02..+0.53$ (never negative)	$\times$
S1 stance order	Δ 0.72, 10/10; agent-field p 0.002	$\checkmark$
S2 teams from stance	8/10 (chance 1.0)	$\checkmark$
S3 off after verdict	-0.01 vs 0.67 during debates	$\checkmark$
#12 within-pair stance (native)	-0.68 (16/18); content ≈ 0	$\checkmark$
#26 / #33 (na- no ballot camps; no unassigned camps)		$\times$ (as predicted)



Second figure. (a) Teammates minus opponents per debate. Content cosine in two embedding models (gray, violet) scatters around zero. Reply stance (orange) is positive in every debate. (b) The same agent pair across re-drafts: stance as opponents minus stance as teammates is negative for 16 of 18 pairs.

GOAL PERIODS IN DETAIL

G12 $\checkmark$ 1b: supported in stance (two-sublattice order; content...
 G26 $\times$ failed
 G33 $\times$ failed

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	2
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Stance labels are zero-shot (DQ2 pair-level “opposes” precision 0.08), so we read only aggregates. The agent-field null assumes replies are independent given agents. Stance measures agreement of positions, so the assigned staggered field alone predicts the order. A coupling beyond the assignment is not shown: content fluctuations stay positively correlated. The sample is seven agents over four days in regime I. #26 has 9 voters with near-identical ballots (low power), and #34 is untouched.

ROUND 2

Test whether stance order needs the protocol: use debates with self-chosen sides or a mid-debate role switch. Fit a reply-level kinetic model to separate a field from a coupling.

- **Essence.** LLM agents don’t form coalitions; assigned roles leave no lasting imprint.
- **H21-R1** Role imprint lifetime: roles bias stance only while active, then reverse (HH127); measure across all role periods.
- **H21-R2** Agreement is the default stance (sycophancy as ferromagnetic stance coupling); disagreement appears only when assigned.

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